MDPI

Article

DATect: Forecasting Domoic Acid Levels from Harmful Algal Blooms along the Pacific Northwest Coast

Anson Chen

Abstract: In spring 2015, the U.S. Pacific Coast experienced an unprecedented spike in domoic acid concentrations, surpassing federal safety limits by over 700% in parts of Washington State and triggering widespread fishery closures. Domoic acid, a potent neurotoxin from harmful algal blooms of Pseudo-Nitzschia, poses significant risks to human health, causing amnesic shellfish poisoning, and is fatal to marine wildlife[1,2]. Previous forecasting methods relied heavily on qualitative approaches, lacking quantitative accuracy, and prior machine learning models were limited in geographic and temporal scope. This project developed a machine learning model to quantify and forecast domoic acid risk across the Pacific Northwest. A comprehensive 21-year database was compiled, incorporating weekly biotic beach sampling data, environmental factors, and satellite observations to train and evaluate various forecasting algorithms. Through rigorous feature selection, key factors influencing domoic acid levels—such as temporal features, Pacific Ocean indices, latitude, and environmental metrics related to harmful algal blooms—were identified. The final model, a random forest regression and classification algorithm called DAtect, achieved a relative mean squared error of 10.57 and an R-squared of 0.57 for direct concentration forecasting. When assessing threat levels against the regulatory limit of 20 ppm, DAtect reached an accuracy of 0.79. This model offers potential to enhance previous qualitative forecasts, enabling proactive measures to protect shellfish fisheries and public health along the Pacific Coast.

Keywords: harmful algal blooms; domoic acid; machine learning; forecasting; Pacific Northwest; environmental monitoring

Citation: Chen, A. DATect: Forecasting Domoic Acid Levels from Harmful Algal Blooms along the Pacific Northwest Coast. *Journal Not Specified* **2024**, *1*, 0. https://doi.org/

Received:
Accepted:
Published:

Copyright: © 2025 by the authors. Submitted to *Journal Not Specified* for possible open access publication under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Background

1.1. Introduction

Domoic acid, a potent neurotoxin produced by certain species of the marine diatom genus *Pseudo-nitzschia*, has become a serious and recurring threat to the Pacific Northwest (PNW) region.[3–6] The blooms of *Pseudo-nitzschia*, known as harmful algal blooms (HABs), carry significant risks for marine ecosystems, public health, and local economies.[6,7] When blooms reach high concentrations, domoic acid can accumulate in shellfish and fish, potentially contaminating seafood on a large scale. Such contamination has necessitated fishing closures and posed ongoing public health risks, making understanding the conditions that drive domoic acid-producing blooms essential for the region.[7–9]

1.2. Environmental Drivers of Pseudo-nitzschia Blooms

The formation of *Pseudo-nitzschia* blooms in the PNW, and the resulting production of domoic acid, are influenced by several key environmental drivers:

Diatoms, including *Pseudo-nitzschia*, require nutrients such as nitrate and silicate for growth. In the PNW, these nutrients become available primarily through upwelling events, which are driven by wind patterns that displace surface waters mostly from the California Current System, allowing nutrient-rich waters from the ocean's depths to rise to the surface.[5,10] This upwelling process, a consistent oceanographic feature along the PNW coast, creates an environment rich in the essential nutrients that promote *Pseudo-nitzschia* proliferation, increasing the likelihood of bloom formation.[9,11,12]

Specific oceanographic features in the PNW further support bloom formation by creating retention zones where nutrient-rich waters and algae can accumulate. The Juan de Fuca eddy near Washington and Heceta Bank off the Oregon coast are two such areas.[11,13] These features promote circulation and retention of *Pseudo-nitzschia* cells, allowing blooms to concentrate and persist over time. In these retention zones, nutrient levels can remain elevated, increasing the likelihood of toxin production by *Pseudo-nitzschia* populations.[6,8,9]

Broader climate patterns, such as El Niño and the Pacific Decadal Oscillation (PDO), play a major role in altering ocean conditions in the Pacific Northwest. For instance, El Niño events often increase sea surface temperatures and disrupt nutrient dynamics by weakening or modifying upwelling.[10,14] The PDO, a long-term ocean-atmosphere phenomenon, shifts between warm and cool phases over periods of decades, influencing water temperature, nutrient distribution, and upwelling intensity in the region.[15] These shifts in oceanographic conditions can impact the frequency, duration, and intensity of Pseudo-nitzschia blooms, creating years with especially high domoic acid production and extended bloom durations.[6,15,16]

A notable example occurred in spring 2015, when a coastwide bloom of the toxigenic diatom Pseudo-nitzschia led to the largest recorded outbreak of domoic acid along the North American west coast.[7] The bloom, driven by anomalously warm ocean conditions, resulted in widespread closures of razor clam, rock crab, and Dungeness crab fisheries, as well as elevated toxin levels in numerous stranded marine mammals.[15] During this event, Pseudo-nitzschia australis thrived north of its usual range in warm, nutrient-poor waters that dominated the northeast Pacific. As upwelling resumed in spring, a series of storms transported nutrient-enriched waters to the coast, fueling the bloom further.[8,16]

1.3. Impact of Domoic Acid Events

The combined environmental factors have led to the development of domoic acid "hotspots" in the PNW, where blooms with high toxin levels can pose immediate risks to marine life and people who consume contaminated seafood.[6,7]

The economic impact of such events has been severe. For example, in 2015, record-breaking domoic acid concentrations led to widespread fishery closures along the U.S. West Coast, with the Dungeness crab fishery alone suffering a 47% revenue decline, amounting to $97.5 million in lost income compared to the previous year.[7] Another major event in 1998 resulted in closures of razor clam and Dungeness crab fisheries in Washington State, causing estimated losses between $15 million and $20 million.[2]

Domoic acid is not only an economic threat but also a public health risk, as it is the primary cause of amnesic shellfish poisoning (ASP). ASP can cause symptoms ranging from gastrointestinal distress to severe neurological damage, including memory loss and, in extreme cases, death. The first documented ASP outbreak occurred in 1987 in Prince Edward Island, Canada, where 107 people fell ill, and three died after consuming contaminated mussels.[5] Since then, similar health threats have persisted along the U.S. Pacific Coast, especially in the PNW where *Pseudo-nitzschia* blooms frequently occur.[3,4]

The toxin also impacts marine wildlife, with notable mass mortality events in sea lions and seabirds. For instance, in Monterey Bay in 1991, domoic acid exposure resulted in the deaths of over 400 California sea lions and numerous seabirds.[3,4] Additional outbreaks have occurred over the years, particularly in 1991, 1998, 2015, and 2017, highlighting the recurring threat that domoic acid poses to the marine ecosystem.[10,15]

1.4. Predictive Modeling and Early Warning Systems

1.4.1. The Pacific Northwest Harmful Algal Bloom Bulletin

The Pacific Northwest Harmful Algal Bloom (PNW HAB) Bulletin is a critical early warning system tailored to forecast the risks associated with domoic acid-producing *Pseudo-Nitzschia* blooms along the coasts of Washington and Oregon.[13] Developed by NOAA in collaboration with the University of Washington and regional shellfish managers, the

Bulletin synthesizes a wide range of environmental data, including *Pseudo-Nitzschia* cell counts, particulate domoic acid levels, and oceanographic conditions such as wind patterns and the position of nutrient-rich upwelling zones, which are key factors driving bloom formation.[10,11,13]

Issued bi-weekly to monthly during peak bloom seasons, the Bulletin is specifically designed to aid shellfish managers in making proactive decisions about shellfish harvesting to protect both the public and the local economy. The Bulletin incorporates environmental indices and real-time data, combining them with historical marine trends and satellite imagery. This data is then analyzed to produce a comprehensive but accessible two-page report that highlights current bloom conditions, estimated risks, and potential changes in domoic acid levels. In particular, it provides managers with a clear picture of bloom risk levels across different coastal regions, helping to prioritize areas where additional sampling or temporary closures may be necessary. By doing so, it supports the management of key fisheries, including those for razor clams and Dungeness crabs, which are economically vital but highly susceptible to domoic acid contamination.[13]

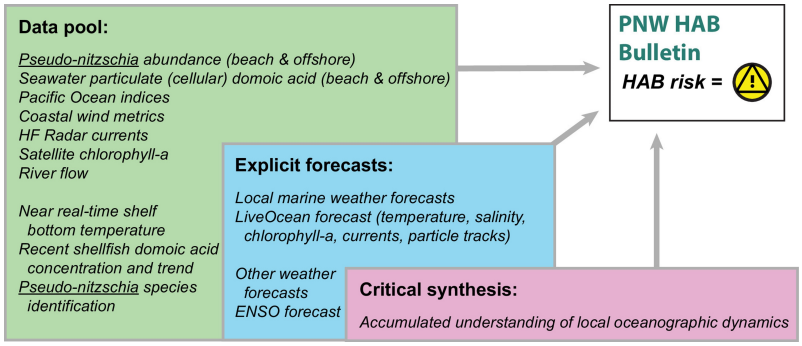

Figure 1. This graphic shows all the data used in the PNW HAB Bulletin to determine regionwide risk [13]

1.4.2. Maine Stakeholder-Based Forecasting System

In coastal Maine, a deep learning forecasting system was developed to provide site-specific predictions of paralytic shellfish toxin (PST) levels, targeting paralytic shellfish poisoning (PSP) events caused by the dinoflagellate *Alexandrium catenella*.[17] This system, developed in partnership with Bigelow Laboratory for Ocean Sciences and the Maine Department of Marine Resources, uses a convolutional neural network (CNN) trained on chemical data collected from routine monitoring sites. The deep learning model forecasts toxicity levels on a weekly basis with high accuracy, achieving over 95% prediction reliability within a one-week timeframe. This approach has transformed forecasting by providing actionable insights at finer spatial and temporal scales, directly benefiting shellfish farmers, harvesters, and public health managers. The weekly forecasts not only predict the onset but also the resolution of PST events, facilitating efficient resource allocation and minimizing economic losses due to closures.[17]

1.4.3. Domoic Acid Forecasting in Northwest Spain

In the Galician Rias Baixas region, machine learning models, particularly Random Forest (RF) and AdaBoost, have been used to forecast *Pseudo-nitzschia* spp. blooms and manage domoic acid risks.[18,19] RF operates by constructing multiple decision trees from randomized data subsets, then "voting" on predictions for higher accuracy and resilience against overfitting, making it well-suited for the region's complex and unbalanced datasets.[18,19] RF's ability to process non-linear relationships has proven effective in capturing the intricate environmental interactions influencing bloom events. Similarly, AdaBoost, an adaptive boosting technique, combines multiple weak classifiers to improve prediction performance by sequentially correcting errors from previous models, which

has increased its effectiveness in scenarios with sparse bloom data by enhancing model sensitivity without overproducing false positives.

Beyond RF and AdaBoost, other machine learning approaches have been explored in Galician HAB forecasting. Support Vector Machines (SVMs) have shown utility in classifying bloom versus no-bloom events by finding the optimal hyperplane that separates bloom data in a high-dimensional space, though these models sometimes fall short in specificity and can generate higher rates of false alarms. Neural Networks (NNs), particularly multilayer perceptrons (MLPs), have been employed to model the non-linear and dynamic nature of bloom data, though they require extensive training data and computational power to achieve high accuracy. These limitations led researchers to explore ensemble methods like RF and AdaBoost, which have shown a more balanced performance in sensitivity and precision for predicting toxic bloom events.[18]

Previous studies also experimented with other configurations to manage the high variability in coastal bloom forecasting. For instance, logistic regression and regression tree models were used for simpler binary classifications but lacked the flexibility required to model complex ecosystem interactions. The high-dimensional and dynamic nature of marine environments in Galicia has therefore driven a shift toward robust machine learning techniques that can process heterogeneous and high-volume data, incorporating varied parameters like upwelling indices, temperature, and salinity. This machine learning approach for domoic acid forecasting has provided more reliable, operational models that are essential for local shellfish harvesting and ASP risk management, demonstrating that ensemble models like RF and AdaBoost offer the most adaptable solutions to the region's forecasting needs.[18]

1.5. Project Aim: Developing a Quantitative Forecasting Model

This project aims to develop a quantitative forecasting model for domoic acid events in the PNW. By analyzing broad environmental data across multiple years and geographic locations, the model seeks to provide one-week advance predictions across a wide range of sites, which could help coastal managers implement proactive strategies to reduce the impact of domoic acid on human health, marine wildlife, and the coastal economy. Enhanced predictive capabilities have the potential to transform the management of HABs in the PNW, improving preparedness and enabling more effective responses to these recurring threats.

2. Methodology

2.1. Data Collection

2.1.1. Biological Data Collection

Data for this project was collected from multiple sources, with careful selection criteria applied to ensure that the final dataset was comprehensive, high-quality, and representative of domoic acid (DA) trends along the Pacific Northwest (PNW) coast. Due to the absence of a centralized database for DA and harmful algal bloom (HAB) monitoring data in the region, historical DA measurements and *Pseudo-nitzschia* cell counts were obtained from state and regional agencies [2,7,13]. Specifically, public records requests were submitted to the Oregon Department of Fish and Wildlife (ODFW), Washington Department of Health (WDOH), and the Olympic Region Harmful Algal Bloom (ORHAB) Partnership. Long-term, site-specific DA measurements and cell counts provided by these agencies formed the backbone of the biological dataset [3,20].

From a pool of over 40 potential monitoring sites along the PNW coast, 10 sites were carefully selected for inclusion in this study. Multiple criteria were applied to select these sites to ensure both data continuity and geographic coverage. Sites were chosen for their comprehensive historical datasets and spatial representation from southern Oregon to northern Washington, minimizing the potential introduction of bias from disproportionate regional representation [2,10]. The selected sites offered consistent sampling data over a multi-year period, which was essential for training a reliable forecasting model. Razor

clams (*Siliqua patula*) were sampled at these sites due to their high susceptibility to DA and low depuration rate, which allows them to retain the toxin longer than other shellfish [21]. This characteristic made razor clams an ideal bioindicator for monitoring DA levels over time, providing a stable and accurate representation of the toxin's persistence in marine ecosystems. Although razor clams provide a reliable bioindicator, potential limitations must be acknowledged; regional variations in DA uptake and retention across different sites may introduce variability in DA measurements, which could affect the generalizability of results beyond the selected sites.

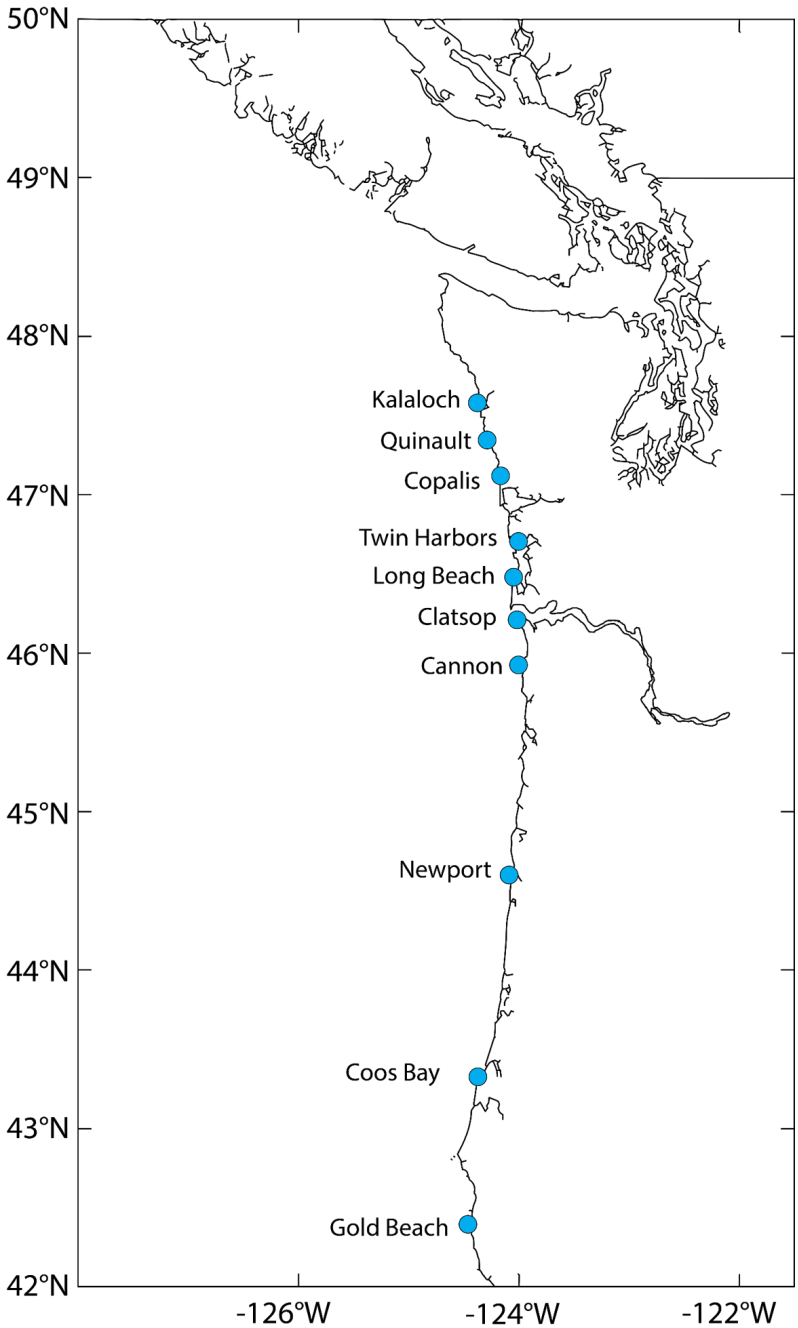

Figure 2. This map shows the location of the 10 forecast sites used in the model.

2.1.2. Temporal Constraints

The initial dataset included records dating back to 1991. However, to align the biological data with available environmental satellite data, the dataset was restricted to the period from 2002 onwards. This decision was informed by the launch of the AQUA-MODIS

satellite, which provided consistent Sea Surface Temperature (SST), Chlorophyll-a (Chl-a), Photosynthetically Available Radiation (PAR), and fluorescence data across the relevant geographic area (49-42°N, -124 to -126°W) [1,22]. Although the Visible Infrared Imaging Radiometer Suite (VIIRS) satellite offers higher accuracy, its operational date in 2015 led to a prioritization of AQUA-MODIS data for its extended temporal coverage, facilitating more comprehensive analysis of DA trends over time [19]. These satellite datasets were downloaded from the NOAA CoastWatch ERDDAP server (links to datasets to be inserted), and 8-day rolling averages were used to mitigate the effects of cloud cover and severe weather, which often obstruct satellite observations in the PNW region.

2.1.3. Environmental Indices

In addition to satellite data, several environmental indices were included to capture broader climatic and oceanographic influences on DA levels. The Biologically Effective Upwelling Transport Index (BEUTI) and Pacific Decadal Oscillation (PDO) were obtained from the NOAA OceanView ERDDAP server [8]. BEUTI estimates vertical nutrient flux into the surface ocean, which can stimulate *Pseudo-nitzschia* blooms, particularly in nutrient-rich upwelling zones along the coast [15]. The PDO captures long-term variability in sea surface temperatures across the North Pacific and has been linked to conditions that favor HAB development [5]. The Oceanic Niño Index (ONI), also from NOAA, provided insights into El Niño and La Niña conditions, which can affect ocean temperatures and nutrient levels in ways that promote DA-producing algal blooms [4].

Finally, data on Columbia River discharge, a critical freshwater input influencing coastal salinity and nutrient dynamics in the PNW, was included. Columbia River flow data was sourced from the US Geological Survey (USGS) server at The Dalles, Oregon, which provided daily discharge measurements [23]. Weekly averages of river flow data were calculated to match the temporal resolution of the other datasets. Each of these environmental variables was selected based on its hypothesized influence on *Pseudo-nitzschia* bloom formation and DA production, allowing the model to account for complex interactions between biological and environmental factors along the PNW coast [24]. Together, these diverse datasets formed a comprehensive foundation for building an effective DA forecasting model.

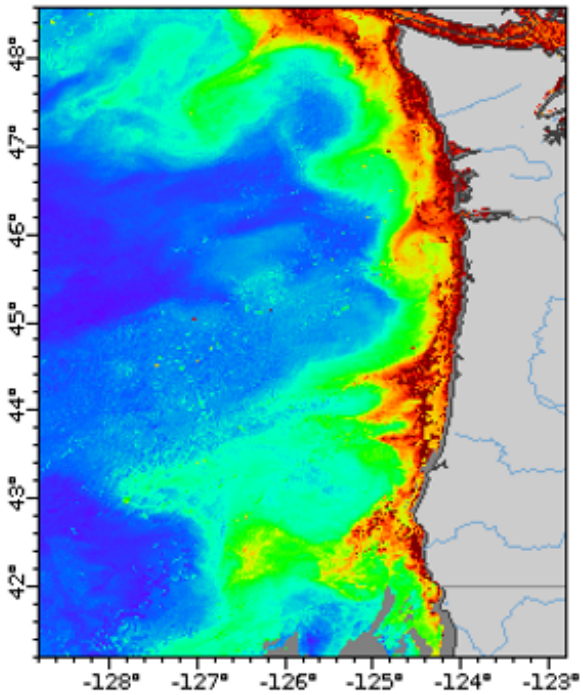

Figure 3. This map shows the satellite Chlorophyll-A levels in 2023 from AQUA-MODIS. Source: NOAA NMFS SWFSC ERD.

2.2. Data Preprocessing

2.2.1. Data Collection and Consolidation

Data was initially sourced from multiple files representing domoic acid and *Pseudo-nitzschia* levels, as well as environmental variables like chlorophyll and sea surface temperature. Each file was loaded individually, with particular attention to the format of collection dates, location identifiers, and variable names [18]. Date formats varied across files, and these were standardized by converting all dates into a "WW-YYYY" format, allowing for consistent weekly aggregation of values across all datasets.

2.2.2. Temporal Alignment and Standardization

To synchronize the data across different temporal resolutions, transformation of the datasets was required. For variables measured at monthly intervals, such as climate indices BEUTI, ONI, and PDO, linear interpolation was applied to derive weekly values, allowing better alignment with the weekly biotic sampling frequency [19]. Daily data, such as Columbia River streamflow, was aggregated into weekly averages to preserve overall trends while minimizing the impact of short-term fluctuations [17]. For satellite-based measurements, such as SST and chlorophyll, 8-day rolling averages were calculated to smooth out daily weather-induced variability [11]. In addition, all dates were converted to Python datetime objects, allowing consistent handling and enabling various datetime-based computations.

2.2.3. Handling Missing Data

Data imputation techniques were applied to handle missing values across the dataset. Median values were used to fill missing values in numerical columns, preventing extreme outliers from overly influencing the imputation process [25]. For environmental data points, such as BEUTI, PDO, and ONI, interpolation techniques were employed, including a K-D tree method to interpolate values across latitude and week-based temporal coordinates [12]. This spatial-temporal interpolation ensured continuous data coverage across both time and space for the environmental indices. To capture the bloom dynamics of *Pseudo-nitzschia*, a

binary matrix was created to represent the presence or absence of blooms across a sliding 1–4 week window. This binary indicator provided temporal insight into bloom progression patterns, offering valuable input for predictive modeling.

2.2.4. Geographic Encoding and Site Identification

Each sampling site was assigned specific latitude and longitude coordinates, enabling spatial analysis in the predictive models. For spatial interpolations, the site coordinates were processed using Ball Tree and K-D Tree structures to support efficient nearest-neighbor queries. This approach was particularly useful for filling missing spatial data points in environmental variables, as spatially aware imputation was achieved based on the proximity of data points in both geographic and temporal dimensions [26]. This step was essential for enabling the models to generalize across different locations along the Pacific Northwest coast.

2.3. Feature Engineering

2.3.1. Spatial Feature Engineering

Spatial clustering was performed using KMeans clustering, assigning each data point to one of five clusters based on latitude and longitude coordinates. These clusters represent distinct spatial areas along the coast, capturing regional similarities in environmental conditions and domoic acid (DA) concentration trends [8]. Each cluster was one-hot encoded, generating binary indicator columns (cluster_0, cluster_1, ..., cluster_4) for use in the model. This spatial clustering approach enhances the model's ability to distinguish between different coastal regions and recognize localized bloom trends [18].

2.3.2. Temporal Feature Engineering

To account for seasonality, cyclic transformations were applied to the day of the year. Using sine and cosine transformations (sin_day_of_year and cos_day_of_year), day-of-year data was represented as a continuous cycle, allowing the model to capture periodic trends in DA concentrations without introducing discontinuities [5]. Additionally, the month of each observation was included as a categorical variable and one-hot encoded, allowing the model to learn monthly seasonal variations. The year was added as a continuous variable to reflect potential long-term trends and interannual variability in bloom patterns [2].

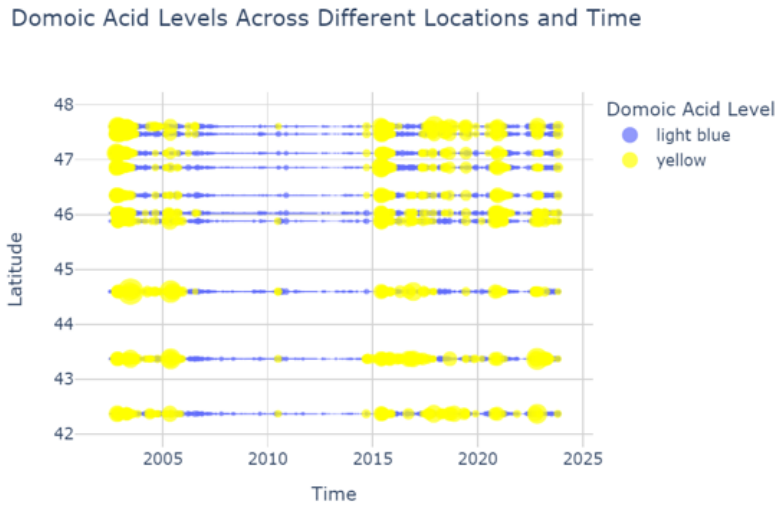

Figure 4. This graph shows trends in Domoic Acid levels over time and latitude, with dot size representing the Domoic Acid concentration. Yellow dots indicate levels above the federal limit of 20 ppm, while blue dots indicate levels below this threshold.

2.3.3. Lag Features

To capture temporal dependencies, lagged versions of the DA level were included for lags of 1, 2, 3, 7, and 14 days. These features allow the model to account for recent changes in DA levels and delayed effects, providing valuable information on bloom persistence and short-term fluctuations. Lagged values were generated per site, maintaining site-specific temporal patterns that might influence DA levels locally [1].

2.3.4. Interaction Terms

For each spatial cluster, interaction terms with seasonal cyclic features were created. By multiplying each cluster indicator (e.g., cluster_0, cluster_1, etc.) with the sin_day_of_year and cos_day_of_year features, interaction terms (cluster_0_sin_day_of_year, cluster_0_cos_day_of_year, etc.) were generated [19]. These interaction terms allow the model to capture how seasonal cycles specifically impact each spatial region, identifying localized seasonal trends in DA levels.

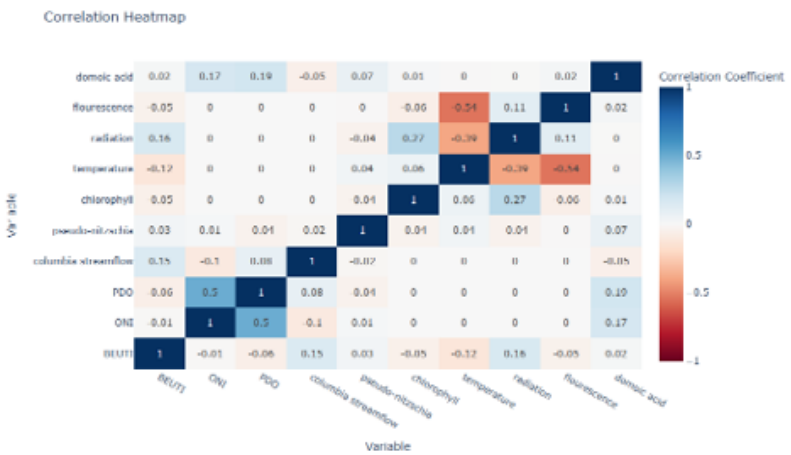

Figure 5. This graphic shows the correlation between various environmental and algal features in the dataset, indicating that, overall, the data exhibits minimal correlation between different features, especially between domoic acid levels and environmental data points such as satellite imagery.

2.3.5. Scaling and Normalization

Continuous features, excluding the categorical and cyclic features, were normalized using MinMax scaling. This scaling process ensured that all continuous variables shared a similar range, preventing features with larger scales from dominating the learning process [23]. Columns excluded from normalization included Date, Site, DA_Levels, and the one-hot encoded cluster indicators.

2.3.6. Feature Selection Subsets

Seven distinct feature subsets were constructed to assess the predictive power of various feature types:

1. All-Inclusive Dataset: Combined all spatial, temporal, and environmental features as a baseline.
2. Spatial-Only Dataset: Focused on spatial data and excluded temporal features to evaluate the predictive power of regional characteristics alone.
3. Temporal-Only Dataset: Included only lagged DA levels and time-based features, isolating the influence of historical trends on DA prediction.
4. Environmental-Only Dataset: Consisted solely of environmental variables to measure the impact of immediate conditions.
5. Cyclic and Interaction Dataset: Focused on the interaction terms between clusters and cyclic seasonal data.

6. Short-Term Lagged Data: Examined only the 1, 2, and 3-day lags for short-term forecasting.
7. Long-Term Lagged Data: Incorporated the 7 and 14-day lags, testing long-term dependencies.

The seven feature selection subsets described were not implemented in the final code. Instead, a comprehensive dataset was used, incorporating all spatial, temporal, and environmental features without partitioning into specific feature subsets [18]. This approach was likely chosen to leverage the full range of available data and maximize the model's predictive power, without performing additional feature subset comparisons.

2.3.7. Feature Encoding for Machine Learning Models

Spatial clusters were one-hot encoded, and month-based categorical variables were binary-encoded. The cyclic encoding of day-of-year using sine and cosine transformations ensured the data captured periodicity in a form compatible with machine learning algorithms [24]. This encoding approach allowed models to recognize seasonal and regional patterns essential for accurate DA forecasting.

2.4. Model Selection

The primary objective of the model selection phase was to identify a predictive model capable of accurately forecasting domoic acid (DA) levels along the Pacific Northwest coast. Initially, a wide range of machine learning and deep learning models was explored, including traditional statistical approaches, ensemble methods, and deep learning architectures. These models were chosen to capture both linear and nonlinear relationships in the data, as well as sequential dependencies that could be present in the time series [19].

One of the challenges encountered during model selection was the complexity of the dataset itself. The compiled database lacked pronounced trends, making it challenging to identify consistent patterns in DA levels [18]. Additionally, the dataset spanned a broad geographic and temporal range, increasing variability and further complicating model training. This lack of strong trends, combined with the sparse and irregular nature of high DA concentration events, presented difficulties for more complex models, especially those requiring extensive computational resources [17]. While deep learning models, such as Feedforward Neural Networks (FFNN), Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN) with Long Short-Term Memory (LSTM), were initially considered due to their capacity to model sequential dependencies, they ultimately proved to be impractical. Training these models was computationally intensive and led to inconsistent results across validation sets due to the limited computing resources available and the lack of strong signal in the data [8].

Moreover, there are currently no quantitative models being used to forecast DA levels specifically for the Pacific Northwest. Previous machine learning applications in HAB research have typically focused on smaller geographic and temporal scopes, limiting their generalizability to a study of this scale [5]. Given these constraints, Random Forest and logistic regression for classification tasks were ultimately compared, as well as Random Forest Regression and multiple linear regression for continuous DA level prediction, all from the Sci-Kit Learn Python library. Random Forest models were selected due to their ability to handle nonlinear relationships and variability in the data, making them well-suited for both regression and classification tasks in a complex environmental dataset [9].

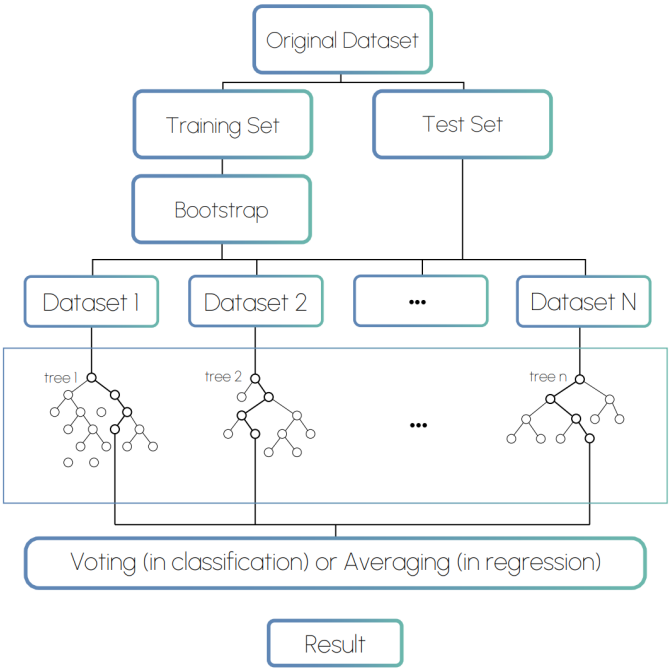

Figure 6. This diagram illustrates the training process of a random forest model for both classification and regression tasks. It shows how multiple decision trees are created from different data subsets, with each tree making predictions independently. In classification, the final prediction is based on the majority vote across trees, while in regression, the prediction is the average output of all trees.

2.5. Model Evaluation and Validation

Model evaluation focused on assessing two primary forecasting tasks: regression-based forecasting of continuous domoic acid (DA) levels and classification-based forecasting of DA risk categories. For regression, the Random Forest model was evaluated based on R^2 and Root Mean Square Error (RMSE), with R^2 representing the proportion of variance in DA levels explained by the model and RMSE quantifying the average prediction error [18].

In contrast, the classification-based evaluation for DA categories relied on accuracy as the primary metric, evaluating the model's ability to predict DA levels relative to regulatory thresholds [23]. By discretizing DA concentrations into predefined risk categories, the classification model provided actionable insights into potential bloom threats, which are crucial for public health and environmental monitoring.

Validation for both regression and classification models was performed using an 80/20 train-test split. Model performance was further compared across different geographic sites, demonstrating the generalizability of the models for both continuous and categorical DA forecasting tasks.

2.5.1. Metrics and Baseline Models

Since no established baseline model existed for DA forecasting of harmful algal blooms in the Pacific Northwest, a base statistical model was implemented for comparative purposes. For continuous DA level forecasting, multiple linear regression was used as the baseline model, while logistic regression served as the baseline for DA category classification [17]. These statistical models provided reference points, enabling a more meaningful assessment of the performance gains achieved by the machine learning approach.

2.5.2. Regression and Classification Metrics

The comparison metrics used for both the Random Forest models and baseline models included R^2, RMSE, and accuracy. For continuous DA level forecasting, R^2 and RMSE were employed to evaluate the models' ability to capture variance and minimize prediction error, respectively [23]. The R^2 metric, computed using `r2_score` from the Sci-Kit Learn Python

library, measures the proportion of variance in the observed data explained by the model, with values ranging from 0 to 1, where 1 indicates a perfect fit [9].

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

where y_i are the observed values, $\hat{y}_i$ are the predicted values, and $\bar{y}$ is the mean of the observed values. RMSE, calculated using mean squared error from the Sci-Kit Learn Python library, represents the square root of the average squared differences between predicted and actual values [23].

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_i = 1^n (y_i - \hat{y}_i)^2}$$

For classification tasks, accuracy was used to measure correct predictions proportionally to total predictions [18]:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

2.6. Deployment and Visualization

In addition to building and evaluating the forecasting models, a deployment and visualization component was developed using the Dash framework. This section was designed to allow interactive exploration of model predictions and site-specific DA trends across the Pacific Northwest coast. The Dash application provided a user-friendly interface allowing users to select between DA level forecasting (regression) and DA category forecasting (classification), view predictions for specific sites, and compare these predictions against actual values over time.[27]

The application layout included drop-down menus for selecting forecast type and geographic site, facilitating user filtering of the displayed data according to their specific needs. Upon selection, the application dynamically updated a line graph displaying both observed and predicted DA values, enabling clear visual comparison of model accuracy over time. For continuous DA level forecasts, the graph displayed DA concentration trends, while for categorical forecasts, it highlighted predicted risk categories. Additionally, performance metrics such as R^2 and RMSE for regression tasks or overall accuracy for classification tasks were shown as annotations, providing users with a quick assessment of model performance.

This interactive visualization served two primary purposes: first, it provided an intuitive platform for researchers and stakeholders to monitor DA trends and assess model accuracy; second, it demonstrated the model's practical applicability for real-time HAB risk assessment. The deployment of this Dash application marked the final stage in the code pipeline, bringing together data preprocessing, model training, evaluation, and visualization into a comprehensive tool for DA forecasting and decision support.

The code for the above can be accessed here: https://github.com/Capanator/DATect-Forecasting-Domoic-Acid

3. Results

3.1. Forecast Accuracy Data

Table 1. Comparison of Forecast Accuracy for Domoic Acid Levels

Continuous Model	**Relative Mean Squared Error**	R^2
DATect (RF Regression)	10.27	0.57
Multiple Linear Regressions	12.70	0.34

Table 2. Comparison of Forecast Accuracy for Domoic Acid Category

Categorical Model	Accuracy
DATect (RF Classification)	0.79
Logistic Regression	0.70
PNW HAB Bulletin	0.917[13]

3.2. Data Forecast Graphs

3.2.1. DA Level Graphs

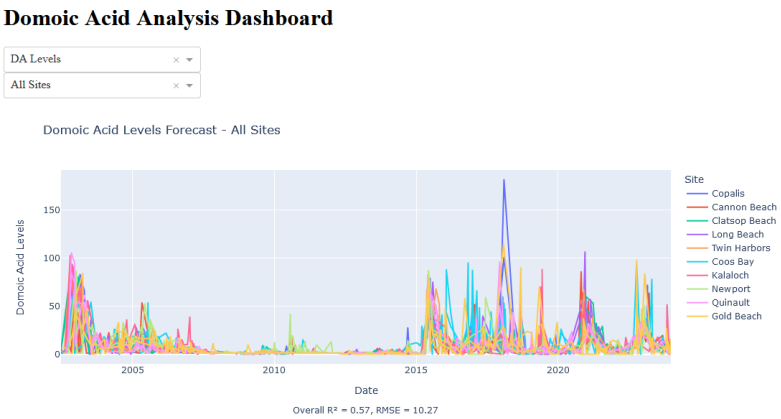

Figure 7. This graph, created using the Plotly and Dash Python libraries, displays the forecasting accuracy for DA levels for all 10 sites

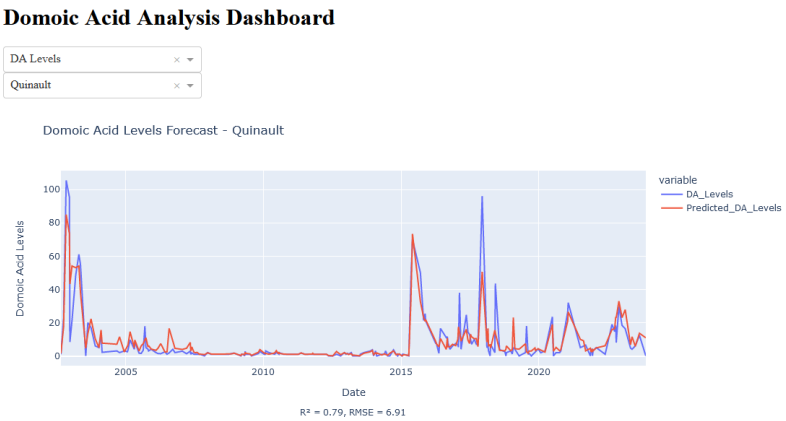

Figure 8. This graph, created using the Plotly and Dash Python libraries, displays the forecasting accuracy for DA levels at the Quinault site. The forecast for this location had the best R^2 and RMSE and is being considered for integration into the PNW HAB Bulletin.

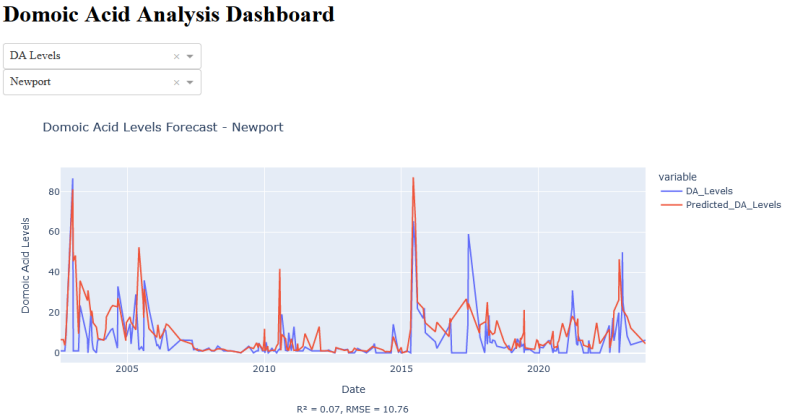

Figure 9. This graph, created using the Plotly and Dash Python libraries, displays the forecasting accuracy for DA levels at the Newport site The forecast for this site had the worst R^2 and RMSE, yet still as seen from the graph, it still is fairly accurate in generally trending towards which direction DA Level is heading

3.2.2. DA Category Graphs

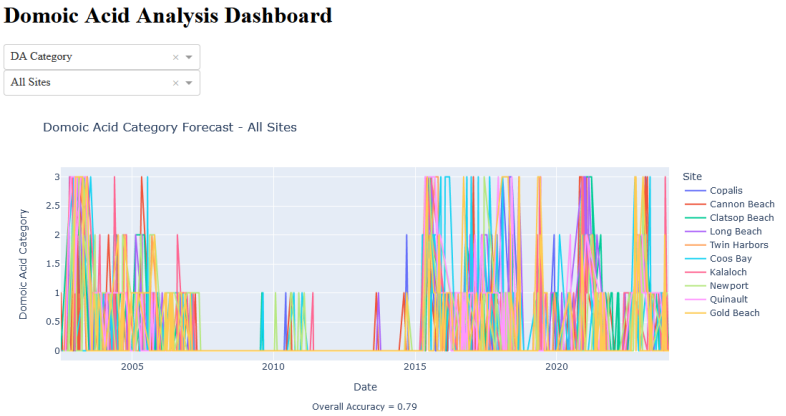

Figure 10. This graph, created using the Plotly and Dash Python libraries, displays the forecasting accuracy for the DA category for all 10 sites

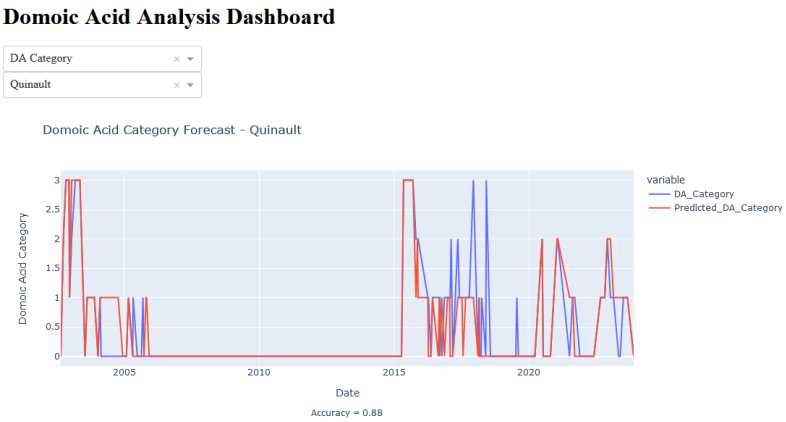

Figure 11. This graph, created using the Plotly and Dash Python libraries, displays the forecasting accuracy for the DA category at the Quinault site. The forecast for this location was the most accurate and is being considered for integration into the PNW HAB Bulletin.

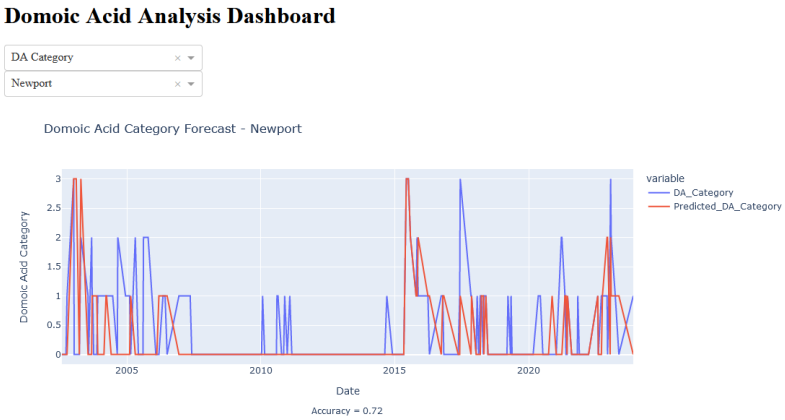

Figure 12. This graph, created using the Plotly and Dash Python libraries, displays the forecasting accuracy for the DA category at the Newport site. Although this site's forecast had the lowest accuracy, it is still reasonably close and remains usable.

4. Discussion

4.1. Analysis of Results

The DAtect model's results underscore its effectiveness in forecasting domoic acid (DA) levels and categories using a Random Forest approach. With an R^2 of 0.57 and an RMSE of 10.27 for regression, as well as an 0.79 classification accuracy, DAtect significantly outperformed traditional statistical baselines, which achieved an R^2 of 0.34, an RMSE of 12.70, and an accuracy of 0.7. The Random Forest model was able to capture complex, non-linear relationships in the environmental data, which other models struggled to address.

Several other machine learning and statistical models were initially tested, including Convolutional Neural Networks (CNNs), Feedforward Neural Networks (FNNs), Long Short-Term Memory networks (LSTMs), Recurrent Neural Networks (RNNs), Support Vector Machines (SVMs), and ARIMA (Auto-Regressive Integrated Moving Average). However, these models faced significant challenges due to the highly uncorrelated nature of the dataset. Unlike classic time-series data, where there are clear dependencies between consecutive observations, the relationships between domoic acid levels and various environmental factors in this dataset are often weak and inconsistent. This lack of strong temporal correlation made it difficult for models like LSTMs and RNNs, which are designed to capture sequential dependencies, to perform well, as they couldn't leverage any meaningful patterns across time steps.

Furthermore, neural network-based models like CNNs, FNNs, and LSTMs were computationally expensive to run, especially given the size and complexity of the dataset. Training these models required significant processing power and time, yet the resulting performance improvements were minimal. CNNs, typically effective in capturing spatial dependencies, were also not a good fit for the dataset, as the data didn't exhibit spatial patterns that CNNs could effectively utilize. FNNs, which rely on dense layers, required careful tuning and often overfitted the data without capturing generalizable patterns, leading to poor results on test data. ARIMA, a statistical model commonly used for time-series forecasting, also struggled with the dataset. The model's reliance on strong autocorrelation in the data limited its effectiveness, as domoic acid levels and the associated environmental factors did not follow predictable trends that ARIMA could leverage. The weak correlations and the presence of multiple influencing factors rendered ARIMA ineffective, as it couldn't capture the complex, multi-variable dynamics involved. Support Vector Machines (SVMs), while powerful for classification in high-dimensional spaces, required extensive computational resources and performed poorly due to the highly variable nature of the data. SVMs struggled to find a hyperplane that could reliably separate DA categories, given the noisy and non-linear relationships among features.

Ultimately, the Random Forest model provided the best balance of computational efficiency and predictive accuracy for this dataset. Random Forests excel at handling uncor-

related features by building an ensemble of decision trees that can pick up on individual relationships within subsets of the data, even if the overall structure is inconsistent. This makes it particularly well-suited for complex environmental datasets like this one, where dependencies are weak, and factors are often not linearly correlated with the target variables. These challenges with other models highlight why Random Forest was ultimately chosen as the backbone of DAtect. It provided a robust solution that could effectively handle the dataset's unique characteristics without requiring the heavy computational load of neural network-based models or the dependency on strong temporal correlations typical of ARIMA and other time-series models.

4.2. DAtect Model Limitations

While DAtect shows promise, it faces several limitations, largely due to the challenges of modeling complex environmental systems. The relationships between environmental factors, such as oceanic indices and DA concentrations, are not straightforward, leading to potential artifacts in the model's predictions. Since the model relies on a Random Forest approach, it can function as a "black box," making it hard to trace the exact reasons behind specific predictions.

The dataset itself also presents challenges. Some data gaps were filled using linear interpolation, which can introduce bias, especially in areas with sparse or inconsistent data. Moving to biweekly data intervals that align with spring tides could help, but may still require interpolation. Using multiple imputation or data from neighboring sites could improve the model's accuracy in handling missing values.

The limitations of satellite-derived data, particularly nearshore, also affect the model's accuracy. Satellite readings in coastal areas are often less reliable due to issues like cloud cover and complex optical properties. Additionally, the current data format in CSV is inefficient for handling large scientific datasets. Transitioning to NetCDF format, which is optimized for multi-dimensional scientific data, could improve data handling, speed up preprocessing, and allow DAtect to scale more effectively.

4.3. DAtect Model Significance

DAtect is a novel tool in harmful algal bloom (HAB) forecasting, offering the first one-week forecast model that predicts a quantitative index for both domoic acid (DA) levels and risk categories. This model stands out not only for its specificity but also for its wide-ranging dataset, which includes over 20 years of historical data from multiple coastal sites along the Pacific Northwest. By integrating diverse sources—shore-based sampling, satellite imagery, and ocean simulations—DAtect captures a broader and more detailed picture than any previous DA models, which often focused on smaller regions or shorter time spans.

As a data-driven model that generates weekly forecasts, DAtect could be a valuable addition to the Pacific Northwest Harmful Algal Bloom (PNW HAB) Bulletin, supplementing its regional, qualitative analysis with precise, site-specific predictions. This local focus could help beach managers and fisheries make better decisions, potentially avoiding unnecessary closures and reducing economic losses, while still prioritizing public health and safety.

DAtect's accuracy in classifying DA categories is also close to that of the PNW HAB Bulletin, showing that machine learning models like this one can be reliable alternatives or complements to expert-driven analysis. Operating independently of human input, DAtect can produce consistent predictions across multiple sites, which could make public health responses and resource management more efficient. The model's ability to confirm correlations between DA levels and larger climate patterns, such as the Pacific Decadal Oscillation (PDO) and Oceanic Niño Index (ONI), suggests it could also be useful in other regions facing similar environmental challenges, broadening its potential impact on global HAB forecasting.

Overall, DAtect's versatility and adaptability could make it an important tool for tracking changes in ocean patterns as climate change progresses. By providing early warnings of HAB risks based on a robust, long-term dataset, DAtect could help coastal communities respond proactively to HAB threats, enhancing both public health protection and environmental management.

4.4. Future Work

To maximize DAtect's potential, several improvements and expansions are planned. One major enhancement would involve converting the data storage format from CSV to NetCDF, which is better suited for handling large-scale scientific datasets with complex structures. This change would improve data processing efficiency, streamline database management, and allow for faster model updates, particularly as more data sources are incorporated.

Expanding DAtect's dataset with real-time, continuous shore-based and offshore Pseudo-nitzschia and DA data could further improve accuracy. This additional data would better align DAtect with satellite spectrometer data and help the model handle "edge cases" where environmental variables are atypical or incomplete. Integrating particle transport models and weather forecasts, like those from the UW LiveOcean model, could improve spatial trend tracking for DA and Pseudo-nitzschia blooms. These additions would enable DAtect to incorporate more comprehensive, context-rich inputs, supporting a better understanding of the factors driving HAB events.

Another area for enhancement is exploring neural network models. Customized neural networks, such as Convolutional Neural Networks (CNNs) or Recurrent Neural Networks (RNNs), could allow DAtect to capture complex temporal and spatial dependencies more effectively. This could be especially beneficial as more continuous datasets become available, making the model more capable of handling nuanced, non-linear trends and improving forecast reliability for extreme DA levels.

Improving DAtect's usability is also a priority. Developing a simple web application would provide beach managers and other stakeholders with easy access to forecasts and real-time DA data. This platform could integrate DAtect with the HAB Bulletin, offering a centralized source of information for managing beach closures and other public health measures. A web interface could also allow users to upload shore-based data, enabling continuous model updates and ensuring forecasts are based on the latest information.

Finally, by training image classification algorithms to recognize HAB-related spectral signatures, DAtect could enhance its ability to monitor algal blooms in nearshore environments, where satellite data is often challenging to interpret. This would enable more accurate assessments of HAB presence and extent, helping coastal managers track blooms more precisely and respond proactively to emerging risks.

Funding: This research received no external funding.

Data Availability Statement: Data will be made available upon reasonable request.

Acknowledgments: The author would like to thank Dr. John Mickett of the University of Washington Applied Plysics Lab for his invaluable help in mentoring this project and helping collect the data needed to create the model.

Conflicts of Interest: The author declares no conflict of interest.

References

1. Moore, S.K.; Mickett, J.B.; Doucette, G.J.; Adams, N.G.; Mikulski, C.M.; Birch, J.M.; Roman, B.; Michel-Hart, N.; Newton, J. An Autonomous Platform for Near Real-Time Surveillance of Harmful Algae and Their Toxins in Dynamic Coastal Shelf Environments. *Journal of Marine Science and Engineering* **2021**, *9*, 336–336. https://doi.org/10.3390/jmse9030336.
2. Trainer, V.L.; King, T.L. SoundToxins: A Research and Monitoring Partnership for Harmful Phytoplankton in Washington State. *Toxins* **2023**, *15*, 189. https://doi.org/10.3390/toxins15030189.
3. Lefebvre, K.A.; Robertson, A. Domoic acid and human exposure risks: A review. *Toxicon* **2010**, *56*, 218–230. https://doi.org/10.1016/j.toxicon.2009.05.034.

4. Trainer, V.L.; Bates, S.S.; Lundholm, N.; Thessen, A.E.; Cochlan, W.P.; Adams, N.G.; Trick, C.G. Pseudo-nitzschia physiological ecology, phylogeny, toxicity, monitoring and impacts on ecosystem health. *Harmful Algae* **2012**, *14*, 271–300. https://doi.org/10.1016/j.hal.2011.10.025.
5. Bates, S.S.; Hubbard, K.A.; Lundholm, N.; Montresor, M.; Leaw, C.P. Pseudo-nitzschia, Nitzschia, and domoic acid: New research since 2011. *Harmful Algae* **2018**, *79*, 3–43. https://doi.org/10.1016/j.hal.2018.06.001.
6. Lewitus, A.J.; Horner, R.A.; Caron, D.A.; Garcia-Mendoza, E.; Hickey, B.M.; Hunter, M.; Huppert, D.D.; Kudela, R.M.; Langlois, G.W.; Largier, J.L.; et al. Harmful algal blooms along the North American west coast region: History, trends, causes, and impacts. *Harmful Algae* **2012**, *19*, 133–159. https://doi.org/10.1016/j.hal.2012.06.009.
7. Moore, S.K.; Dreyer, S.J.; Ekstrom, J.A.; Moore, K.; Norman, K.; Klinger, T.; Allison, E.H.; Jardine, S.L. Harmful algal blooms and coastal communities: Socioeconomic impacts and actions taken to cope with the 2015 U.S. West Coast domoic acid event. *Harmful Algae* **2020**, *96*, 101799. https://doi.org/10.1016/j.hal.2020.101799.
8. Trainer, V.L.; Kudela, R.M.; Hunter, M.V.; Adams, N.G.; McCabe, R.M. Climate Extreme Seeds a New Domoic Acid Hotspot on the US West Coast. *Frontiers in Climate* **2020**, *2*. https://doi.org/10.3389/fclim.2020.571836.
9. Moreno, A.R.; Anderson, C.; Kudela, R.M.; Sutula, M.; Edwards, C.; Bianchi, D. Development, calibration, and evaluation of a model of Pseudo-nitzschia and domoic acid production for regional ocean modeling studies. *Harmful Algae* **2022**, *118*, 102296. https://doi.org/10.1016/j.hal.2022.102296.
10. McKibben, S.M.; Peterson, W.; Wood, A.M.; Trainer, V.L.; Hunter, M.; White, A.E. Climatic regulation of the neurotoxin domoic acid. *Proceedings of the National Academy of Sciences* **2017**, *114*, 239–244. https://doi.org/10.1073/pnas.1606798114.
11. Trainer, V.L.; Hickey, B.M.; Lessard, E.J.; Cochlan, W.P.; Trick, C.G.; Wells, M.L.; MacFadyen, A.; Moore, S.K. Variability of Pseudo-nitzschiaand domoic acid in the Juan de Fuca eddy region and its adjacent shelves. *Limnology and Oceanography* **2009**, *54*, 289–308. https://doi.org/10.4319/lo.2009.54.1.0289.
12. González Vilas, L.; Spyrakos, E.; Pazos, Y.; Torres Palenzuela, J.M. A New Algorithm Using Support Vector Machines to Detect and Monitor Bloom-Forming Pseudo-nitzschia from OLCI Data. *Remote Sensing* **2024**, *16*, 298. https://doi.org/10.3390/rs16020298.
13. McCabe, R.M.; Hickey, B.M.; Trainer, V.L. The Pacific Northwest Harmful Algal Blooms Bulletin. *Harmful Algae* **2023**, *127*, 102480. https://doi.org/10.1016/j.hal.2023.102480.
14. Zhu, Z.; Qu, P.; Fu, F.; Tennenbaum, N.; Tatters, A.O.; Hutchins, D.A. Understanding the blob bloom: Warming increases toxicity and abundance of the harmful bloom diatom Pseudo-nitzschia in California coastal waters. *Harmful Algae* **2017**, *67*, 36–43. https://doi.org/10.1016/j.hal.2017.06.004.
15. McCabe, R.M.; Hickey, B.M.; Kudela, R.M.; Lefebvre, K.A.; Adams, N.G.; Bill, B.D.; Gulland, F.M.D.; Thomson, R.E.; Cochlan, W.P.; Trainer, V.L. An unprecedented coastwide toxic algal bloom linked to anomalous ocean conditions. *Geophysical Research Letters* **2016**, *43*. https://doi.org/10.1002/2016gl070023.
16. Sauvey, A.; Claquin, P.; Roy, B.L.; Jolly, O.; Fauchot, J. Physiological conditions favorable to domoic acid production by three Pseudo-nitzschia species. *Journal of Experimental Marine Biology and Ecology* **2023**, *559*, 151851–151851. https://doi.org/10.1016/j.jembe.2022.151851.
17. Grasso, I.; Archer, S.D.; Burnell, C.; Tupper, B.; Rauschenberg, C.; Kanwit, K.; Record, N.R. The hunt for red tides: Deep learning algorithm forecasts shellfish toxicity at site scales in coastal Maine. *Ecosphere* **2019**, *10*. https://doi.org/10.1002/ecs2.2960.
18. Aláez, F.M.B.; Palenzuela, J.M.T.; Spyrakos, E.; Vilas, L.G. Machine Learning Methods Applied to the Prediction of Pseudo-nitzschia spp. Blooms in the Galician Rias Baixas (NW Spain). *ISPRS International Journal of Geo-Information* **2021**, *10*, 199. https://doi.org/10.3390/ijgi10040199.
19. Cruz, R.C.; Reis Costa, P.; Vinga, S.; Krippahl, L.; Lopes, M.B. A Review of Recent Machine Learning Advances for Forecasting Harmful Algal Blooms and Shellfish Contamination. *Journal of Marine Science and Engineering* **2021**, *9*, 283. https://doi.org/10.3390/jmse9030283.
20. Mos, L. Domoic acid: a fascinating marine toxin. *Environmental Toxicology and Pharmacology* **2001**, *9*, 79–85. https://doi.org/10.1016/s1382-6689(00)00065-x.
21. Schreiber, S.; Hanisak, M.D.; Perricone, C.S.; Fonnegra, A.C.; Sullivan, J.; McFarland, M. Pseudo-nitzschia species, toxicity, and dynamics in the southern Indian River Lagoon, FL. *Harmful Algae* **2023**, *126*, 102437. https://doi.org/10.1016/j.hal.2023.102437.
22. Mckibben, S.; Watkins-Brandt, K.; Wood, A.; Hunter, M.; Forster, Z.; Hopkins, A.; Du, X.; Eberhart, B.T.; Peterson, W.; White, A. Monitoring Oregon Coastal Harmful Algae: Observations and implications of a harmful algal bloom-monitoring project. https://doi.org/10.1016/j.hal.2015.10.004.
23. Yu, P.; Gao, R.; Zhang, D.; Liu, Z.P. Predicting coastal algal blooms with environmental factors by machine learning methods. *Ecological Indicators* **2021**, *123*, 107334. https://doi.org/10.1016/j.ecolind.2020.107334.
24. Manning, N.; Wang, Y.C.; Long, C.; Bertani, I.; Sayers, M.; Bosse, K.; Shuchman, R.; Scavia, D. Extending the forecast model: Predicting Western Lake Erie harmful algal blooms at multiple spatial scales. *Journal of Great Lakes Research* **2019**. https://doi.org/10.1016/j.jglr.2019.03.004.
25. Cusack, C.; Mouriño, H.; Moita, M.T.; Silke, J. Modelling Pseudo-nitzschia events off southwest Ireland. *Journal of Sea Research* **2015**, *105*, 30–41. https://doi.org/10.1016/j.seares.2015.06.012.
26. Clark, S.; Hubbard, K.A.; McGillicuddy, D.J.; Ralston, D.K.; Shankar, S. Investigating Pseudo-nitzschia australis introduction to the Gulf of Maine with observations and models. *Continental Shelf Research* **2021**, *228*, 104493. https://doi.org/10.1016/j.csr.2021.104493.

27. Garcia-Robledo, A.; Zangiabady, M. Dash Sylvereye: A Python Library for Dashboard-Driven Visualization of Large Street Networks. *IEEE access* **2023**, *11*, 121142–121161. https://doi.org/10.1109/access.2023.3327008.